Evidence-Traceable, Status-Aware Gene–Disease Relation Extraction with a Large Language Model: A Multi-Benchmark Evaluation

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Abstract

Objective: To develop and evaluate a zero-shot method that extracts gene–disease relations from biomedical papers together with sentence-level evidence, a four-way status (supported, uncertain, explicit negative, not found), and a source label (this study or background), addressing the evidence-free, positive-only output of current extraction tools.

Methods: A large language model (Claude Opus 5.5) reads each paper in a single request and follows a frozen one-page decision card; a deterministic validator checks every cited span against the source text. Without benchmark-specific tuning, we evaluated the method on BioRED (419 abstracts), EU-ADR (100 abstracts), and a 40-article full-text sample of RENET2, with document-level pair matching, bootstrap confidence intervals, and PubTator3 as the comparator.

Results: On BioRED, precision was 0.894 (95% CI 0.855–0.931) and recall 0.430. An operating point chosen on Train+Dev raised held-out Test F1 from 0.542 to 0.627 with precision unchanged. PubTator3, which is trained on BioRED, scored higher on a 50-abstract test sample (F1 0.726 vs 0.551), and the two systems together recovered 73% of gold relations. On EU-ADR, F1 was comparable (0.506 vs 0.529) and all three negative gold pairs were labelled explicit negative. On RENET2 full text, our recall exceeded PubTator3's (0.370 vs 0.254), and only 2 of 223 supported pairs contradicted a gold label. Re-querying 97 unjudged same-sentence pairs yielded no new supported relation. Replacing the model with GPT-6.1 Sol gave the same F1 (0.555). Cost was USD 0.056 per abstract and USD 0.44 per full text.

Conclusion: A written decision card turns a general-purpose LLM into a precise, auditable gene–disease extractor that complements supervised tools. Its recall gap mainly reflects a stricter evidence standard than benchmark annotators apply, and the status labels let users adjust this trade-off.

Keywords: Relation Extraction; Gene–Disease Association; Large Language Models; Biocuration; Evidence Traceability

1. Introduction

Gene–disease associations underpin variant interpretation, drug-target discovery, and curated knowledge bases such as DisGeNET [1] and CTD [2]. Most of this knowledge is still published as free text in journal articles. Curators read each paper, decide whether it supports an association, and record the evidence. This work does not scale with a literature that grows by more than a million articles a year.

Curation also needs more than a list of related pairs. A curator must see which sentences support a relation, so the claim can be checked in seconds. A paper may support a relation, hedge it, or report that no association was found, and collapsing these into "related" or "not related" discards negative and speculative findings. Curators also distinguish a paper's own results from its citations of prior work, because the two carry different weight.

Automatic relation extraction has improved steadily on benchmarks such as BioRED [3]. Supervised encoders fine-tuned on annotated corpora [4, 5, 6] and literature-scale systems such as PubTator3 [7] now reach high F1 in their training domain. Their output, however, is a set of positive pairs without evidence, status, or source. Most are trained on abstracts, whereas in our 100-paper full-text corpus 90% of supported relations had evidence only outside the abstract.

Large language models (LLMs) can read full papers and follow written instructions, which makes them candidates for this richer task. Free-form prompting, however, produces outputs that are hard to verify and vary between runs [8, 9]. Benchmark comparisons are also hard to interpret when gold standards are incomplete, so an apparent false positive may be a valid relation that annotators did not label [9, 10].

This study writes the annotation guideline as a compact decision card, requires a fixed line-based output that cites sentence IDs, and lets a deterministic validator reject any answer that cites text not present in the paper. We evaluate the frozen method on three benchmarks with different annotation standards, compare it with PubTator3, and analyse where it disagrees with each gold standard. We find that the method is consistently precise and rarely contradicts explicit negative labels, and that its recall gap reflects a stricter evidence standard rather than overlooked text. Choosing which uncertain pairs to count, using a rule fixed on development data, raised held-out BioRED Test F1 from 0.542 to 0.627 at unchanged precision.

2. Related Work

2.1. Supervised Biomedical Relation Extraction

Domain-pretrained encoders such as BioBERT [4] and PubMedBERT [5], fine-tuned on annotated corpora, remain the standard approach. BioREx [6] improved BioRED results by pooling heterogeneous corpora during training, and subsequent work added relation directionality [11]. PubTator3 [7] applies such a model, trained on BioRED [3], together with entity recognition and normalization over PubMed and PMC. It is the most widely used public resource for this task and serves as our main comparator. These systems predict a relation label for each entity pair and are trained mostly on abstracts.

2.2. Large Language Models for Relation Extraction and Biocuration

Recent benchmarks show that zero- and few-shot LLMs lag behind fine-tuned encoders on biomedical relation extraction [8, 12, 13, 14]. In the BioCreative VIII BioRED track, community systems relied primarily on fine-tuned encoders, while direct LLM prompting struggled with recall and schema adherence [12]. End-to-end zero-shot evaluations on BioRED [13] and across 61 biomedical triplet datasets [14] found that general-purpose LLMs often miss implicit relations or output unaligned entity spans. To make LLM output usable for curation, SPIRES (OntoGPT) [15] fills schema templates and grounds terms to ontologies, and DRAGON-AI [16] combines retrieval-augmented generation with ontology curation. Most of these studies use short prompts and score only positive predictions. Our work differs in three ways: a written decision procedure instead of a brief prompt, machine-checked evidence spans, and explicit scoring of uncertain and negative statuses.

2.3. Gene–Disease Benchmarks and Incomplete Gold Standards

BioRED [3] annotates eight relation types between concept pairs in 600 abstracts and marks whether each relation is a new finding (*Novel*). EU-ADR [17] labels gene–disease pairs in abstracts as positive, speculative, or negative by annotator consensus. RENET2 [18] provides document-level labels on full texts, with negative labels assigned automatically by tools. Evaluating generative extractors against such corpora is difficult because valid relations are often unannotated. Wadhwa et al. [9] showed through manual review that many apparent LLM false positives are valid relations missed by annotators, and Fan et al. [10] separated unannotated valid extractions from factual errors. Crowdsourced re-annotation of EU-ADR found that single-annotator relations and

guideline ambiguities account for many disagreements [19]. We therefore audit contradictions of explicit negative labels separately from unlabelled extra predictions.

3. Methods

3.1. Datasets

Table 1 summarizes the three benchmarks.

3.1.1. BioRED

We used all 419 BioRED abstracts that contain at least one gene–disease relation (Train 275, Dev 71, Test 73), with 1,633 gold relations. A relation counts when one endpoint is a gene or gene product and the other a disease or phenotypic feature, regardless of relation subtype. A 50-abstract Test sample was used for the head-to-head comparison with PubTator3. Rules for counting uncertain pairs were chosen on Train+Dev only, and Test was held out.

3.1.2. EU-ADR

We used the 100 abstracts of the EU-ADR target–disorder subset. Its 134 consensus pairs carry graded labels: 122 positive (PA), 9 speculative (SA), and 3 negative (NA). Unless stated otherwise, PA pairs are the gold relations.

3.1.3. RENET2

We sampled 40 full-text articles from RENET2 (seed 20261028). The released texts contain the title, abstract and early body. The sample has 181 associated, 83 non-associated and 1 ambiguous pair; non-associated labels were assigned by tools and not checked by hand. Three associated pairs involve an entity absent from the released text; we keep them in the recall denominator.

3.1.4. Full-text corpus

To test full-text behaviour without gold labels, we ran the method on 100 heterogeneous PMC articles (research articles, case reports, reviews, and one conference-abstract collection) and compared the output with PubTator3.

Table 1: Benchmarks used in this study.

Benchmark	Text	Documents used	Gold gene–disease relations	Labels beyond "related"
BioRED [3]	PubMed abstracts	419 abstracts with $\geq$1 gene–disease relation (Train 275 / Dev 71 / Test 73); 50-abstract Test sample	1,633	Novel / not novel
EU-ADR [17]	PubMed abstracts	Target–disorder subset, 100 abstracts	134 consensus pairs: 122 PA, 9 SA, 3 NA	Positive / speculative / negative
RENET2 [18]	PMC full text (title, abstract, early body)	Random sample of 40 articles	181 associated; 83 non-associated; 1 ambiguous	Non-associated labels tool-assigned

3.2. Relation Extraction

3.2.1. Task Definition

The input is one paper as text. The output is a set of gene and disease entities, each linked to the surface forms that occur verbatim in the text, and one of four statuses for every gene $\times$ disease pair (Table 2). Supported pairs also carry a relation type (causal, correlation, or association) and a source label (this study, background, both, or unclear). Symptoms and general phenotypes are recorded as auxiliary entities rather than disease endpoints, which affects evaluation (Section 3.3).

Table 2: The four pair statuses.

Status	Definition
`supported`	The text reports a finding or an explicit attribution linking the pair (not an aim or a hypothesis), and the link is readable in the text.
`uncertain`	There is potential evidence, but hedging, scope, identity, co-reference, presupposition or a conflict prevents a decision. A reason code is required.
`explicit_negative`	The text explicitly reports no association, or an opposite result, for this pair on a stated question.
`not_found`	Everything else, including co-occurrence only. It does not mean the pair is medically unrelated.

3.2.2. Decision Card

The decision rules began as a detailed annotation guideline refined on development papers before any benchmark run. We condensed it into a one-page decision card of about 1,000 words, paired with an output format of about 600 words. The card defines entity typing and merging, the four statuses, relation types, source labels, and key boundary judgments: co-occurrence alone is never a relation; the authors' own speculation is `uncertain` (`hedged_claim`); a variant detected without disease attribution is not a relation; and statistical tests apply to a pair only when they test that gene against that disease outcome. The card was frozen as variant V1 before the benchmark experiments and was not edited for any benchmark.

Before freezing, we compared three variants on a fixed, stratified set of 12 full-text papers never used to edit the rules. Condensing the full guideline into the card (V1 vs V2) did not reduce quality: pair-status agreement was 0.976 and the first request was about 12% cheaper. Low reasoning effort (V3) reduced cost by 24% but lost about 10% of genes, diseases and supported relations, so we kept medium effort. [TODO: add a minimal-prompt baseline without the card.]

3.2.3. Single-Request Pipeline

The pipeline runs in four steps (Fig. 1). (1) *Packing:* the paper is split into blocks and sentences with stable IDs (e.g. `b0012.s3`) and the reference list is removed; the card and format form a cached system prompt. (2) *Generation:* one request to Claude Opus 5.5 with medium reasoning effort and all tools disabled returns pipe-separated lines for entities, evidence, pairs, study findings, conflicts and rule gaps; evidence is cited by sentence ID only. (3) *Expansion:* a script locates every surface form in the text and builds the mention table and full gene × disease matrix, with unlisted pairs set to `not_found`. (4) *Validation:* the validator checks that every surface form occurs verbatim, every sentence ID exists, pair endpoints are primary entities, and uncertain pairs carry valid reason codes. A failed check triggers one repair request listing the errors. On the 100-paper corpus, 69 papers passed on the first request and 31 needed one repair; all 100 passed, and all repairs fixed formatting rather than content. Fig. 2 shows the output for one paper.

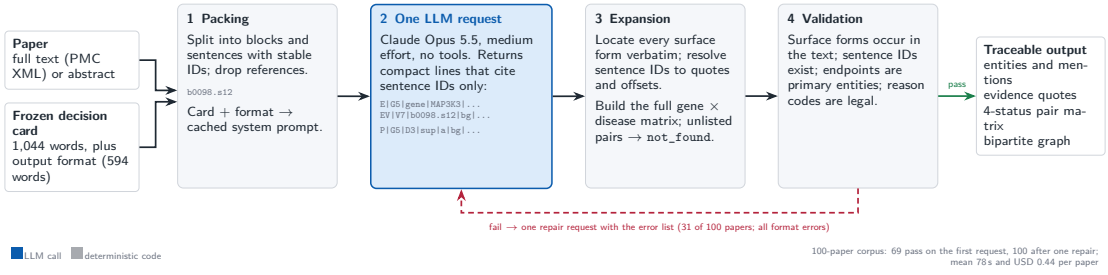


Figure 1: The single-request pipeline. Only step 2 calls the LLM; packing, expansion and validation are deterministic code. A paper that fails validation gets one repair request listing the errors.

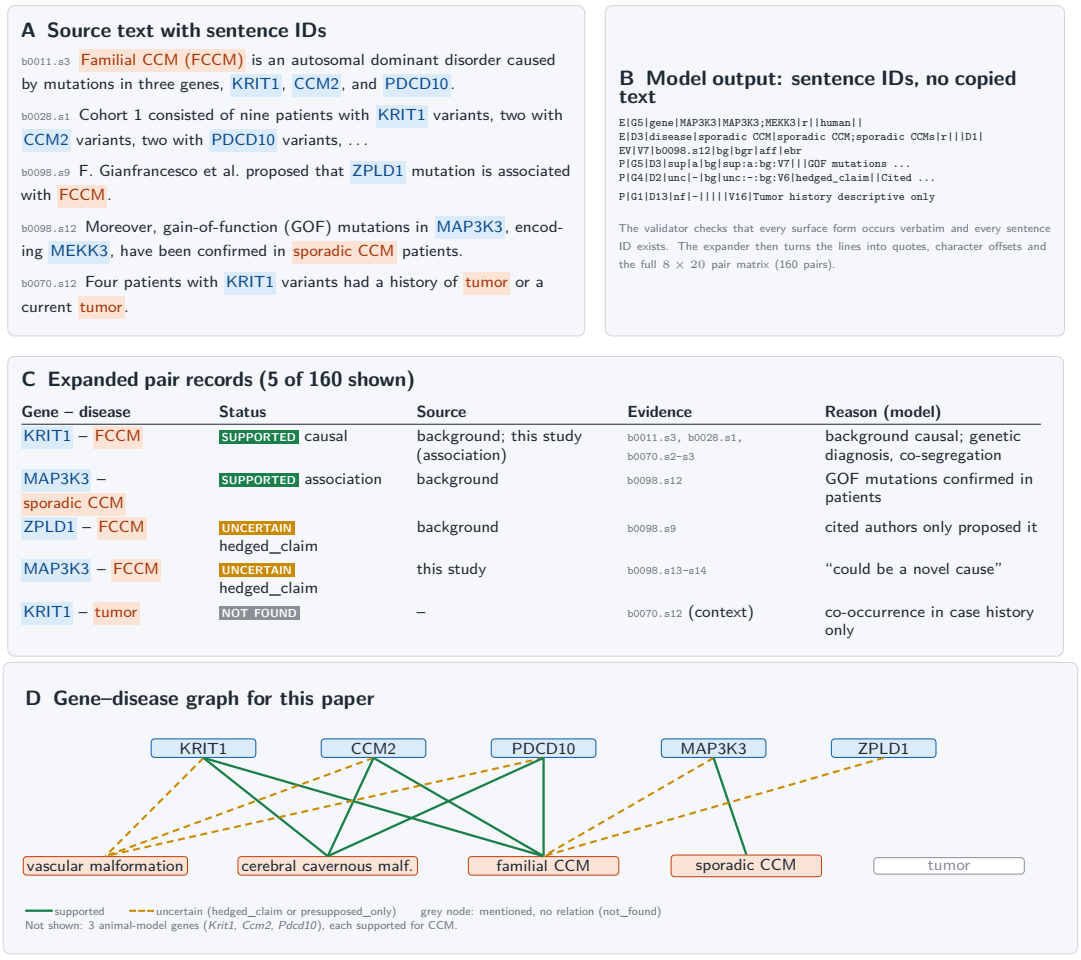


Figure 2: Traceable extraction on one full-text article, PMC10014009 [20]. (A) Excerpts with stable sentence IDs; genes in blue, disease mentions in coral. (B) The model's compact output cites sentence IDs only. (C) Deterministically expanded pair records covering three statuses and both sources. (D) The resulting per-paper bipartite graph. Quotes are reproduced from an open-access article (CC BY-NC-ND 4.0) for illustration.

3.2.4. Comparators

PubTator3 [7]: relation predictions from the public API. Its relations come from abstracts only, even for the RENET2 full texts, and its relation extractor is trained on BioRED, so BioRED is in-domain for it. *GPT-6.1 Sol*: the same frozen card, format, input text and validator, run via the Codex CLI, to test whether results depend on the model. [TODO: add an open-weight LLM with the same card, and published supervised BioRED results as reference rows.]

3.3. Evaluation

The evaluation unit is one (gene concept, disease concept) pair per document. A predicted entity aligns with a gold concept when any of its mentions overlaps a gold mention by character span within the same text block, so spelling variants align without database normalization. We report precision, recall and F1, with 95% confidence intervals from a bootstrap over documents (2,000 resamples, seed 20261028) [21], because pairs within a document are not independent. The same script scores all benchmarks and systems.

The statuses support several operating points from one run: *strict* (only `supported` pairs count as predictions; the default), *supported + uncertain* (all uncertain pairs also count), and *supported + hedged/scope* (uncertain pairs count only when their reason codes are restricted

to `hedged_claim` and/or `scope`; chosen on BioRED Train+Dev and evaluated once on Test). Separately, the *scope-adjusted view* removes from the recall denominator gold pairs whose disease our rules classify as a symptom or phenotype. We report it alongside the strict view, never instead of it.

4. Results

4.1. Performance Comparison

Table 3 and Fig. 3 summarize performance. On all 419 BioRED abstracts, 89% of `supported` pairs matched gold relations (precision 0.894, recall 0.430). Results were consistent across splits, with F1 of 0.584 on Train, 0.608 on Dev and 0.542 on Test, and the scope-adjusted view raised recall to 0.468 (F1 0.614).

On the 50-abstract BioRED Test sample, PubTator3 achieved similar precision (0.827 vs 0.835) and higher recall (0.647 vs 0.412). The two systems were complementary: of 238 gold relations, both found 78, only PubTator3 found 76, only our method found 20, and both missed 64, so together they recovered 174 (73%). On EU-ADR, F1 was similar (0.506 vs 0.529, overlapping intervals), with our method at higher precision (0.469 vs 0.408) and PubTator3 at higher recall (0.754 vs 0.549). On the 40 RENET2 full texts, our method achieved higher recall (0.370 vs 0.254; 0.464 when counting `uncertain` pairs), because PubTator3 extracts relations from abstracts only.

Replacing Claude Opus 5.5 with GPT-6.1 Sol under the same card gave F1 0.555 (vs 0.551) on the BioRED sample, and pair-status agreement between the two models on 11 full-text papers was 0.986.

Table 3: Main results, strict view unless stated. Brackets: 95% bootstrap confidence intervals over documents.

Benchmark (documents)	System / operating point	Precision	Recall	F1 [95% CI]
BioRED, all (419)	Ours	0.894 [0.855, 0.931]	0.430	0.581 [0.538, 0.625]
	Ours, supported + uncertain	0.814	0.577	0.676 [0.636, 0.714]
BioRED Test (73)	Ours	0.853	0.397	0.542 [0.444, 0.644]
	Ours, supported + hedged/scope	0.846	0.498	0.627
BioRED Test sample (50)	Ours	0.835	0.412	0.551 [0.440, 0.665]
	Ours with GPT-6.1 Sol	0.853	0.412	0.555 [0.465, 0.639]
	PubTator3	0.827	**0.647**	**0.726** [0.609, 0.825]
EU-ADR, PA (100)	Ours	**0.469** [0.350, 0.600]	0.549	0.506 [0.403, 0.602]
	PubTator3	0.408 [0.326, 0.498]	**0.754**	0.529 [0.447, 0.612]
RENET2 full text (40)	Ours	0.318 [0.206, 0.470]	0.370	0.342 [0.258, 0.421]
	Ours, supported + uncertain	0.312	**0.464**	0.373
	PubTator3	**0.623** [0.511, 0.750]	0.254	0.361 [0.243, 0.475]

4.2. Status Labels Against Graded Gold Labels

EU-ADR's graded labels test the statuses that BioRED cannot. All 3 gold negative (NA) pairs were labelled `explicit_negative` by our method, whereas PubTator3 predicted one of them as a relation. Of the 122 positive (PA) pairs, our method labelled 67 `supported`, 16 `uncertain`, 10 `not_found` and 4 `explicit_negative`; for 25, an endpoint was not recorded as a primary entity. Counting uncertain pairs raised PA recall to 0.680 at precision 0.414. On RENET2, only 2 of our 223 `supported` pairs carried the silver non-associated label. On BioRED, our source label agreed with the *Novel* label for 526 of 703 matched relations (75%).

The reason codes attached to `uncertain` pairs also separated likely relations from unlikely ones (Fig. 4B). On BioRED Train+Dev, the share of uncertain pairs that matched gold relations was 82% for `hedged_claim` only ($n = 99$), 75% for `hedged_claim` + `scope` ($n = 32$), 62% for `scope` only ($n = 113$), 38% for `presupposed_only`, and 6% for `identity`. Counting only the first three groups as predictions raised held-out Test F1 from 0.542 to 0.627: recall rose from 0.397 to 0.498 while precision went from 0.853 to 0.846, with no extra model calls.

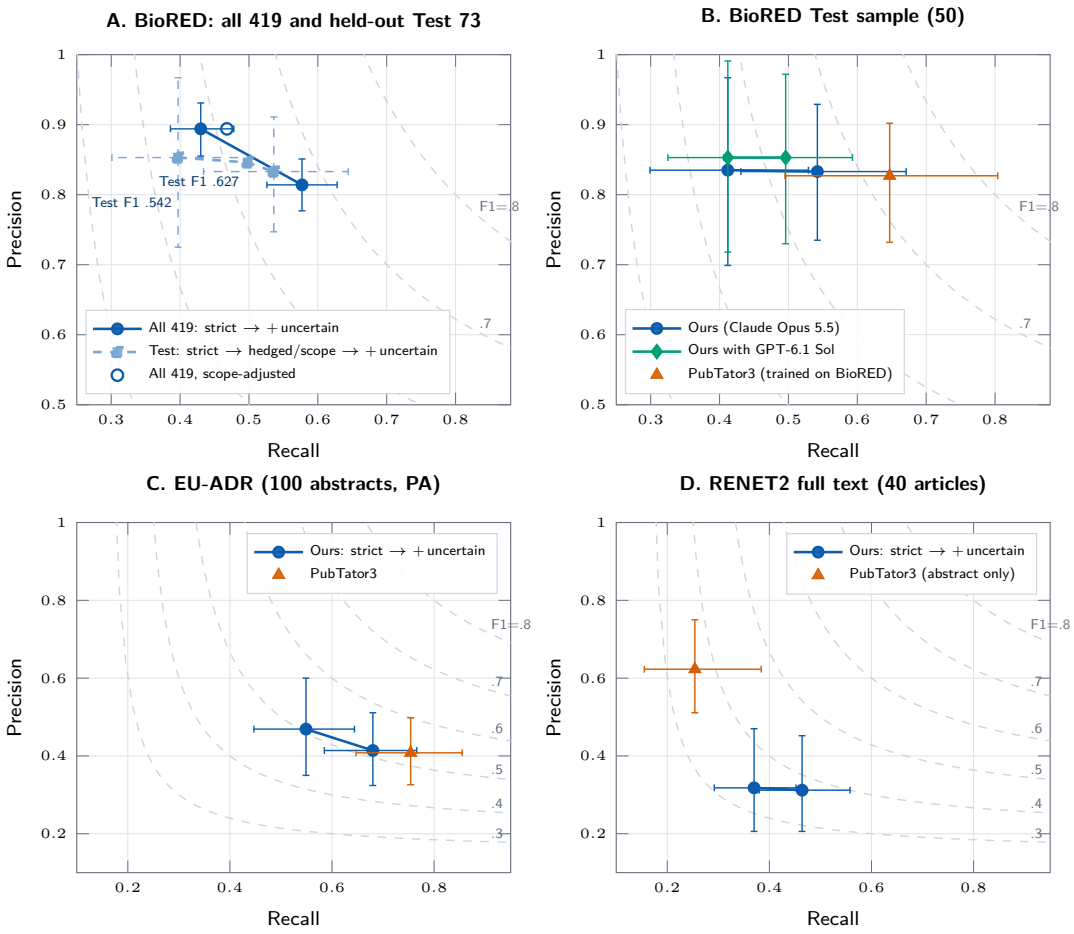


Figure 3: Precision and recall of each system and operating point, with 95% bootstrap confidence intervals over documents. Dashed grey curves are F1 contours from 0.3 to 0.8 in steps of 0.1; panels A and B are zoomed to precision ≥ 0.5. Connected points are operating points of one run: strict (supported only) and adding uncertain pairs; in panel A, the middle Test point is the hedged/scope operating point chosen on Train+Dev (no interval computed). On EU-ADR and RENET2, measured precision is a lower bound because the gold sets are incomplete.

4.3. Error Analysis

4.3.1. Missed Relations

Most missed BioRED relations were pairs the model read and judged rather than overlooked (Fig. 4A). Of the 930 missed gold relations, 547 (59%) were judged but not supported: 295 `not_found`, 240 `uncertain`, and 12 `explicit_negative`. The other 383 reflect entity scope: 131 disease endpoints kept as symptoms or phenotypes, and 159 diseases and 93 genes not recorded as entities. On Train+Dev, most unrecorded entities were gene families and complexes (e.g. NF-κB, MAPK, Wnt, NMDA receptors), which the card excludes as single-gene endpoints, and generic disease terms such as "cancer", "tumor" and "inflammation". Even if every such pair were recorded and judged supported, recall would rise by at most 4.0 and 6.6 percentage points.

To test whether the model overlooked co-mentioned pairs, we ran two experiments on the Dev split (71 abstracts). On Train+Dev, 117 of the 155 unjudged gold relations (75%) had the gene and disease in the same sentence. First, variant V4 required an explicit judgment for every co-mentioned pair; recall fell from 0.454 to 0.403, precision rose from 0.922 to 0.947, supported pairs dropped from 153 to 133, and cost rose by 16%. Second, a script listed all 97 unjudged same-sentence pairs in 18 Dev abstracts and a second request asked the model to judge each one; 92 remained `not_found`, 4 became `uncertain`, 1 became `explicit_negative`, and none became

`supported`, leaving all metrics unchanged at an extra cost of USD 0.64. When asked directly, the model still judged that the text did not meet the card's criteria, so the recall gap reflects a stricter evidence threshold than BioRED annotators used rather than an attention failure.

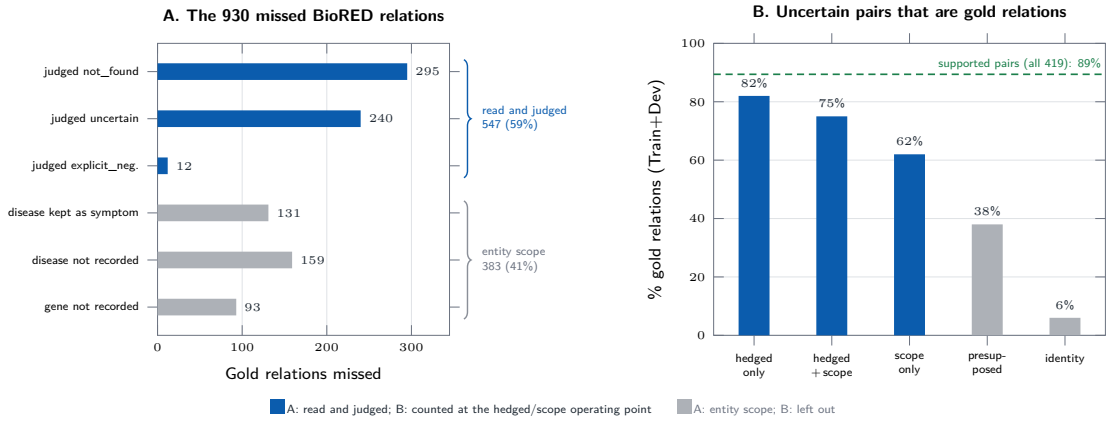


Figure 4: (A) The 930 BioRED gold relations missed in the strict view (all 419 abstracts); most were read and judged by the model rather than overlooked. (B) Share of `uncertain` pairs on Train+Dev that are gold relations, by reason code. Pairs with only `hedged_claim` and/or `scope` reasons (blue) are counted at the hedged/scope operating point, which raised held-out Test F1 from 0.542 to 0.627.

4.3.2. Extra Predictions

The 87 BioRED `supported` pairs counted as false positives fell into three groups: for 60, BioRED annotated both concepts but no relation; for 23, our disease endpoint (such as a clinical outcome) was not a BioRED disease concept; and for 4, our gene did not match a BioRED gene concept. On EU-ADR, 10 of our 37 extra predictions involving gold concepts had been marked by a single annotator and therefore fell outside the consensus set. On RENET2, of our 152 supported pairs outside the gold set, 105 had never been labelled because no tool had proposed them, 45 involved an entity that is not a RENET2 concept, and only 2 carried the non-associated label. Of the 152, 136 were relations we labelled as background; measured precision was 0.53 for relations labelled this study and 0.28 for background relations, consistent with RENET2's focus on each article's main findings.

4.4. Full-Text Extraction and Cost

Across the 100 PMC full-text papers, our method extracted 972 `supported` relations, 451 `uncertain` pairs and 47 `explicit_negative` pairs, whereas PubTator3 reported 198 relations. Of the 972 supported relations, 875 (90%) were supported only by sentences outside the abstract. Twenty-five papers yielded no supported relation: 8 contained no gene, 8 no disease, and 9 only co-occurrences, diagnostic panel listings, or speculative statements. At list API prices, including repairs, cost was USD 0.056 per BioRED abstract, USD 4.75 for the 100 EU-ADR abstracts, USD 6.36 for the 40 RENET2 articles, and USD 0.44 per full-text paper, with a mean model time of 78 s. [TODO: annotate 15–20 of these papers to measure full-text precision and recall.]

5. Discussion

5.1. Why the Benchmarks Disagree

The same frozen pipeline scored strict F1 of 0.58 on BioRED, 0.51 on EU-ADR, and 0.34 on RENET2. Because no rule or prompt changed across benchmarks, the spread reflects how each benchmark defines a relation. BioRED annotates relations stated in abstracts, including hedged associations, and groups symptoms, phenotypes and broad terms such as "cancer" under its

disease category. EU-ADR keeps only relations on which at least two annotators agreed. RENET2 labels only candidate pairs proposed by its tools and assigns negative labels automatically. A precision-oriented extractor is therefore penalized differently by each protocol, and on EU-ADR and RENET2 its measured precision is a lower bound [9, 10]. Across the corpora with negative labels, the method rarely contradicted an explicit negative annotation: none of the 3 EU-ADR negative pairs and 2 of 223 supported RENET2 pairs. Most apparent false positives are unlabelled pairs rather than contradictions, which is the property that matters when a curator verifies the cited sentence.

5.2. Implications for Curation

Four-way statuses let one extraction run serve different workflows. A database pipeline can import `supported` pairs automatically and route `uncertain` pairs tagged `hedged_claim` or `scope` to human review; on BioRED Test, this second group added 10 recall points at almost no precision cost. Binary classifiers cannot separate hedged claims from entity-identity errors without retraining. The method also complements supervised tools: PubTator3 achieves higher recall within its training domain, whereas our method captures full-text evidence, distinguishes study findings from background citations, and identifies explicit negatives. Their union recovered 73% of BioRED sample relations against 65% for PubTator3 alone, which suggests a hybrid in which PubTator3 proposes candidate pairs and the decision card judges each with evidence. Because the card is human-readable and version-controlled, adapting the method to another relation type, such as chemical–disease associations, means editing a page of rules rather than annotating a training corpus. Near-identical results with GPT-6.1 Sol suggest that the behaviour is carried by the written rules rather than by one model.

5.3. Limitations

First, we have not yet manually adjudicated the disputed pairs; our reading of extra predictions as unannotated relations rests on each benchmark's label structure and spot checks rather than dual-annotator review. Second, the RENET2 evaluation used 40 articles and the PubTator3 comparison on BioRED used 50 abstracts, so several confidence intervals are wide, and we did not run paired significance tests. Third, entities are aligned by character span and not mapped to NCBI Gene or MeSH, which prevents aggregation across documents and direct comparison with DisGeNET [1] or CTD [2]. Fourth, Claude Opus 5.5 and GPT-6.1 Sol are proprietary models that may change over time; we archive all prompts, raw responses and scoring scripts. Fifth, the benchmark abstracts are public and may have been seen in pretraining, as BioRED was used to train PubTator3. Sixth, each configuration was run once; agreement of 0.976–0.985 between variants on 12 papers suggests low run-to-run variation, but we did not measure it on the full benchmarks. Finally, the 100-paper corpus has no gold labels, and RENET2's released texts are truncated.

6. Conclusions

A frozen, one-page decision card enabled a general-purpose LLM to extract gene–disease relations from full-text papers with machine-checked sentence evidence, four-way statuses and source labels at USD 0.44 per paper. Across three benchmarks with distinct annotation protocols, the method achieved high precision on BioRED (0.894) and rarely contradicted explicit negative labels on EU-ADR or RENET2. Re-querying unjudged same-sentence pairs yielded no additional supported relations, indicating that the recall gap reflects a stricter evidence standard rather than missed text, and an operating point chosen on development data recovered 10 recall points on held-out data. Future work will add dual-annotator adjudication of disputed pairs, normalization to NCBI Gene and MeSH, and hybrid pipelines that pair supervised candidate generation with rule-guided LLM verification.

CRediT authorship contribution statement

[TODO: to be agreed with co-authors]

Declaration of competing interest

[TODO: to complete]

Data and code availability

Code, the decision card, prompts, raw model outputs and scoring scripts will be released at [TODO: repository URL]. EU-ADR and RENET2 texts are not redistributed; download scripts are provided.

Declaration of generative AI use

[TODO: Required by the publisher. Describe the use of AI tools in drafting this manuscript; check the current Elsevier wording.]

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